

Cefpar SB Injection 1 g
Cefoperazone 500 mg
Sulbactam 500 mg

Product Name

Cefpar SB Injection 1 g

Product Description

Off white crystalline powder.

A clear and colourless solution after reconstituted.

Composition

Each vial contains:

Cefoperazone sodium equivalent to Cefoperazone 500 mg

Sulbactam sodium equivalent to Sulbactam 500 mg

Pharmacodynamics

The antibacterial component is cefoperazone, a 3rd generation cephalosporin, which acts against sensitive organisms during the stage of active multiplication by inhibiting biosynthesis of cell wall mucopeptide. Sulbactam does not possess any useful antibacterial activity, except against Neisseriaceae and Acinetobacter. However, biochemical studies with cell-free bacterial systems have shown it to be an irreversible inhibitor of most important β -lactamases produced by β -lactam antibiotic-resistant organisms. The potential for sulbactam preventing the destruction of penicillins and cephalosporins by resistant organisms was confirmed in whole-organism studies using resistant strains in which sulbactam exhibited marked synergy with penicillins and cephalosporins. As sulbactam also binds with some penicillin-binding proteins, sensitive strains are also often rendered more susceptible to sulbactam/cefoperazone than to cefoperazone alone.

The combination of sulbactam and cefoperazone is active against all organisms sensitive to cefoperazone. In addition, it demonstrates synergistic activity (up to 4-fold reduction in minimum inhibitory concentrations for the combination versus those for each component) in a variety of organisms, most markedly the following: *Haemophilus influenzae*, *Bacteroides* sp, *Staphylococcus* sp, *Acinetobacter calcoaceticus*, *Enterobacter aerogenes*, *Escherichia coli*, *Proteus mirabilis*, *Klebsiella pneumoniae*, *Morganella morganii*, *Citrobacter freundii*, *Enterobacter cloacae*, *Citrobacter diversus*.

Sulbactam/cefoperazone is active in vitro against a wide variety of clinically significant organisms: Gram-Positive Organisms: *Staphylococcus aureus* (penicillinase- and non-penicillinase-producing strains), *Staphylococcus epidermidis*, *Streptococcus pneumoniae* (formerly *Diplococcus pneumoniae*), *Streptococcus pyogenes* (Group A β - hemolytic streptococci), *Streptococcus agalactiae* (Group B β -hemolytic streptococci), most other strains of β -hemolytic streptococci, many strains of *Streptococcus faecalis* (enterococcus).

Gram-Negative Organisms: *Escherichia coli*, *Klebsiella* sp, *Enterobacter* sp. and *Citrobacter* sp, *Haemophilus influenzae*, *Proteus mirabilis*, *Proteus vulgaris*, *Morganella morganii* (formerly *Proteus morganii*), *Providencia rettgeri* (formerly *Proteus rettgeri*), *Providencia* sp, *Serratia* sp (including *S. marcescens*), *Salmonella* sp, *Shigella* sp, *Pseudomonas aeruginosa* and some other *Pseudomonas* sp, *Acinetobacter calcoaceticus*, *Neisseria gonorrhoeae*, *Neisseria meningitidis*, *Bordetella pertussis* and *Yersinia enterocolitica*.

Anaerobic Organisms: Gram-negative bacilli (including *Bacteroides fragilis*, other *Bacteroides* sp, and *Fusobacterium* sp). Gram-positive and gram-negative cocci (including *Peptococcus*, *Peptostreptococcus* and *Veillonella* sp).

Gram-positive bacilli (including *Clostridium*, *Eubacterium* and *Lactobacillus* sp).

Pharmacokinetics

Approximately 84% of the sulbactam dose and 25% of the cefoperazone dose administered with sulbactam/cefoperazone is excreted by the kidney. Most of the remaining dose of cefoperazone is excreted in the bile. After sulbactam/cefoperazone administration, the mean half-life for sulbactam is about 1 hr while that for cefoperazone is 1.7 hrs. Serum concentrations have been shown to be proportional to the dose administered. These values are consistent with previously published values for the agents when given alone. Mean peak sulbactam and cefoperazone concentrations after the administration of 2 g of sulbactam/cefoperazone (1 g sulbactam, 1 g cefoperazone) IV over 5 min were 130.2 and 236.8 mcg/mL, respectively. This reflects the larger volume of distribution for sulbactam ($V_d=18-27.6$ L) compared to cefoperazone ($V_d=10.2-11.3$ L).

After IM administration of 1.5 g sulbactam/cefoperazone (0.5 g sulbactam, 1 g cefoperazone) peak serum concentrations of sulbactam and cefoperazone are seen from 15 min to 2 hrs after administration. Mean peak serum concentrations were 19 and 64.2 mcg/mL for sulbactam and cefoperazone, respectively.

After multiple dosing, no significant changes in the pharmacokinetics of either component of sulbactam/cefoperazone have been reported and no accumulation has been observed when administered every 8-12hrs.

Renal Dysfunction: In patients with different degrees of renal function administered sulbactam/cefoperazone, the total body clearance of sulbactam was highly correlated with estimated creatinine clearance. Patients, who are functionally anephric, show a significantly longer half-life of sulbactam (mean 6.9 and 9.7 hrs in separate studies). Hemodialysis significantly altered the half-life, total body clearance and volume of distribution of sulbactam. No significant differences have been observed in the pharmacokinetics of cefoperazone in renal failure patients.

Elderly: The pharmacokinetics of sulbactam/cefoperazone have been studied in elderly individuals with renal insufficiency and compromised hepatic function. Both sulbactam and cefoperazone exhibited longer half-life, lower clearance and larger volumes of distribution when compared to data from normal volunteers. The pharmacokinetics of sulbactam correlated well with the degree of renal dysfunction while for cefoperazone there was a good correlation with the degree of hepatic dysfunction.

Children: Studies conducted in paediatrics have shown no significant changes in the pharmacokinetics of the components of Cefoperazone Sulbactam compared to adult values. The mean half-life in children has ranged from 0.91-1.42 hrs for sulbactam and from 1.44-1.88 hrs for cefoperazone.

Indication

Monotherapy: Treatment of the following infections when caused by susceptible organisms: Respiratory tract infections (upper and lower); urinary tract infections (upper and lower); peritonitis, cholecystitis, cholangitis and other intra-abdominal infections; septicaemia; meningitis; skin and soft tissue infections; bone and joint infections; pelvic inflammatory disease, endometritis, gonorrhoea and other infections of the genital tract.

Combination Therapy: Because of the broad spectrum of activity of sulbactam/cefoperazone, most infections can be treated adequately with this antibiotic alone. However, Cefoperazone Sulbactam may be used concomitantly with other antibiotics if such combinations are indicated.

If an aminoglycoside is used, renal function should be monitored during the course of therapy.

Recommended Dosage

Adults:

Daily dosage recommendations for sulbactam/cefoperazone in adults are as follows:

Ratio	SBT/CPZ(g)	Sulbactam activity (g)	Cefoperazone activity (g)
1:1	2.0 - 4.0	1.0 - 2.0	1.0 – 2.0

Doses should be administered every 12 hrs in equally divided doses.

In severe or refractory infections the daily dosage of sulbactam/cefoperazone may be increased up to 8 g (ie, 4 g cefoperazone activity). Patients may require additional cefoperazone administered separately. Doses should be administered every 12 hrs in equally divided doses. The recommended maximum daily dosage of sulbactam is 4 g.

Renal Dysfunction:

Dosage regimens of sulbactam/cefoperazone should be adjusted in patients with marked decrease in renal function (creatinine clearance of <30 mL/min) to compensate for the reduced clearance of sulbactam. Patients with creatinine clearances between 15 and 30 mL/min should receive a maximum of 1 g of sulbactam administered every 12 hrs (maximum daily dosage of 2 g sulbactam), while patients with creatinine clearances of <15 mL/min should receive a maximum of 500 mg of sulbactam every 12 hrs (maximum daily dosage of 1 g sulbactam). In severe infections it may be necessary to administer additional cefoperazone.

The pharmacokinetic profile of sulbactam is significantly altered by hemodialysis. The serum half-life of cefoperazone is reduced slightly during hemodialysis. Thus, dosing should be scheduled to follow a dialysis period.

Children:

Daily dosage recommendations for sulbactam/cefoperazone in children are as follows:

Ratio	SBT/CPZ (mg/kg/day)	Sulbactam activity (mg/kg/day)	Cefoperazone activity (mg/kg/day)
1:1	40 – 80	20 – 40	20 – 40

Doses should be administered every 6 to 12 hours in equally divided doses. In serious or refractory infections, these dosages may be increased up to 160 mg/kg/day. Doses should be administered in 2-4 equally divided doses.

Neonates:

For neonates in the 1st week of life, the drug should be given every 12 hrs. The maximum daily dosage of sulbactam in paediatrics should not exceed 80 mg/kg/day. For doses of sulbactam/cefoperazone requiring >80 mg/kg/day cefoperazone activity, additional Cefoperazone should be administered separately.

Route of Administration:

Intramuscular:

Lidocaine HCl 2% is a suitable vehicle for Intramuscular administration but not for initial reconstitution (see **Incompatibilities** and **Reconstitutions**).

Intravenous:

For intermittent infusion, each vial of sulbactam/cefoperazone should be reconstituted with the appropriate amount (see **Reconstitutions**) followed by administration over 15 to 60 minutes.

Lactated Ringer's Solution is a suitable vehicle for intravenous infusion, however, not for initial reconstitution (see **Incompatibilities** and **Reconstitutions**). For intravenous injection, each vial should be reconstituted (see **Reconstitutions**) and administered over a minimum of 3 minutes.

Reconstitutions

Reconstitute one vial Cefoperazone/Sulbactam containing 1.0g as follows:

Route of administration	Diluent	Volume of diluent	Final Concentration (mg/ml)	Temperature	Use Period
Intramuscular	Sterile Water for Injections + Lidocaine HCl 2% solution	2.4 mL + 1 mL	125 + 125	25°C	24 Hours
Intravenous	Sterile Water for Injections + Sterile Water for Injections	4 mL + 20 mL	25 + 25	25°C	24 Hours
	5% dextrose in water + 5% dextrose in water	4 mL + 20 mL	25 + 25	25°C	24 Hours
	0.9% Sodium Chloride Injection + 0.9% Sodium Chloride Injection	4 mL + 20 mL	25 + 25	25°C	24 Hours
Intravenous (With Ringer's lactate solution)	Sterile Water for Injections & Ringer's lactate solution	4 mL + 100 mL	5 + 5	25°C	24 Hours

Incompatibilities

Aminoglycosides:

Solutions of sulbactam/cefoperazone and aminoglycosides should not be directly mixed, since there is a physical incompatibility between them.

If combination therapy with sulbactam/cefoperazone and an aminoglycoside is contemplated (see **Indications**) this can be accomplished by sequential intermittent IV infusion provided that separate secondary IV tubing is used, and that the primary IV tubing is adequately irrigated with an approved diluent between doses. It is also suggested that doses of sulbactam/cefoperazone be administered throughout the day at times as far removed from administration of the aminoglycoside as possible.

Lactated Ringer's Solution:

Initial reconstitution with Lactated Ringer's Solution should be avoided since these mixtures have been shown to be incompatible. However, a 2- step dilution process involving initial reconstitution in water for injection will result in a compatible mixture when further diluted with Lactated Ringer's Solution (see **Dosage & Administration and Reconstitution**).

Lidocaine:

Initial reconstitution with 2% lidocaine HCl solution should be avoided since this mixture has been shown to be incompatible.

However, a 2-step dilution process involving initial reconstitution in water for injection will result in a compatible mixture when further diluted with 2% lidocaine HCl (see **Reconstitution**).

Contraindications

Cefoperazone Sulbactam is contraindicated in patients with known allergy to penicillins, sulbactam, cefoperazone or any of the cephalosporins. If however allergy occurs, the drug should be discontinued and the appropriate therapy instituted.

Warnings and Precautions

General: As with other antibiotics, vitamin K deficiency has occurred in a few patients treated with cefoperazone. The mechanism is most probably related to the suppression of gut flora which normally synthesizes this vitamin. Those at risk include patients with poor diet, malabsorption states (e.g., cystic fibrosis) and patients on prolonged IV alimentation regimens. Prothrombin time should be monitored in these patients receiving anticoagulant therapy and exogenous vitamin K administered as indicated.

As with other antibiotics, overgrowth of non-susceptible organisms may occur during prolonged use of sulbactam/cefoperazone. Patients should be observed carefully during treatment. As with any potent systemic agent, it is advisable to check periodically for organ system dysfunction during extended therapy; this includes renal, hepatic and hematopoietic systems. This is particularly important in neonates, especially when premature and other infants.

Serious and occasionally fatal hypersensitivity (anaphylactic) reactions have been reported in patients receiving β -lactam or cephalosporin therapy. These reactions are more apt to occur in individuals with a history of hypersensitivity reactions to multiple allergens. If an allergic reaction occurs, the drug should be discontinued and the appropriate therapy instituted.

Serious anaphylactic reactions require immediate emergency treatment with epinephrine. Oxygen, IV steroids and airway management, including intubation, should be administered as indicated.

Hepatic Dysfunction: Cefoperazone is extensively excreted in bile. The serum half-life of cefoperazone is usually prolonged and urinary excretion of the drug increased in patients with hepatic diseases and/or biliary obstruction. Even with severe hepatic dysfunction, therapeutic concentrations of cefoperazone are obtained in bile and only a 2- to 4-fold increase in half-life is seen.

Dose modification may be necessary in case of severe biliary obstruction, severe hepatic disease or in cases of renal dysfunction coexistent with either of those conditions.

In patients with hepatic dysfunction and concomitant renal impairment, cefoperazone serum concentrations should be monitored and dosage adjusted as necessary. In these cases, dosage should not exceed 2 g/day of cefoperazone without close monitoring of serum concentrations.

Cefoperazone does not displace bilirubin from plasma protein-binding sites.

Effects on the Ability to Drive or Operate Machinery: Clinical experience with sulbactam/cefoperazone indicates that it is unlikely to impair a patient's ability to drive or use machinery.

Use in Infancy: Cefoperazone/Sulbactam has been effectively used in infancy. It has not been extensively studied in premature infants or neonates.

Therefore, in treating premature infants and neonates potential benefits and possible risks involved should be considered before instituting therapy.

Interactions with Other Medicaments

Alcohol: A reaction characterized by flushing, sweating, headache and tachycardia has been reported when alcohol was ingested during and as late as the 5th day after cefoperazone administration. A similar reaction has been reported with certain other cephalosporins and patients should be cautioned concerning ingestion of alcoholic beverages in conjunction with administration of sulbactam/cefoperazone. For patients requiring artificial feeding orally or parenterally, solutions containing ethanol should be avoided.

Drug-Laboratory Test Interactions: A false-positive reaction for glucose in the urine may occur with Benedict's or Fehling's solution.

Pregnancy and Lactation

Pregnancy:

Reproduction studies have been performed in rats at doses up to 10 times the human dose and have revealed no evidence of impaired fertility and no teratological findings. Sulbactam and cefoperazone cross the placental barrier. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, this drug should be used during pregnancy only if clearly needed.

Nursing Mothers:

Only low concentrations of sulbactam and cefoperazone are excreted in human milk. Although both drugs pass poorly into breast milk of nursing mothers, caution should be exercised when sulbactam/cefoperazone is administered to a nursing mother.

Side Effects

Sulbactam/cefoperazone is generally well tolerated. The majorities of adverse events are of mild or moderate severity and are tolerated with continued treatment. In pooled clinical trial data from comparative and non-comparative studies to approximately 2500 patients, the following was observed:

Gastrointestinal:

As with other antibiotics, the most frequent side effects observed with sulbactam/cefoperazone have been gastrointestinal. Diarrhoea/loose stools 3.9% have been reported most frequently followed by nausea and vomiting 0.6%.

Dermatologic Reactions:

As with all penicillins and cephalosporins, hypersensitivity manifested by maculopapular rash, urticaria 0.08% has been reported. These reactions are more likely to occur in patients with a history of allergies, particularly to penicillin.

Hematology:

Slight decreases in neutrophils have been reported. As with other β -lactam antibiotics, reversible neutropenia may occur with prolonged administration. Some individuals have developed a positive direct Coombs' test during treatment. Decreased haemoglobin 0.9% (13/1416) or hematocrits 0.9% (13/1409) have been reported, which is consistent with published literature on cephalosporins. Transient eosinophilia 3.5% (40/1130) and thrombocytopenia 0.8% (11/1414) have occurred and hypoprothrombinemia 3.8% (10/262) has been reported.

Miscellaneous:

Headache 0.04%, fever 0.5%, injection pain 0.08% and chills 0.04% occurred in <1% of patients.

Laboratory Abnormalities:

Transient elevations of liver function tests, SGOT 5.7% (94/1638), SGPT 6.2% (95/1529), alkaline phosphatase 2.4% (37/1518) and bilirubin 1.2% (12/1040) levels have been noted.

Local Reactions:

Sulbactam/cefoperazone is well tolerated following IM administration. Occasionally, transient pain may follow administration by this route. As with other cephalosporins and penicillins, when sulbactam/cefoperazone is administered by an IV catheter some patients may develop phlebitis at the infusion site.

In post-marketing experience, the following additional undesirable effects have been reported:

General:

Anaphylactoid reaction (including shock).

Cardiovascular: Hypotension.

Gastrointestinal: Pseudomembranous colitis.

Hematopoietic: Leukopenia.

Skin/Appendages: Pruritus, Stevens-Johnson syndrome.

Urinary: Hematuria.

Vascular: Vasculitis.

Symptoms and Treatment of Overdose

Limited information is available on the acute toxicity of cefoperazone sodium and sulbactam sodium in humans. Overdosage of the drug would be expected to produce manifestations that are principally extensions of the adverse reactions reported with the drug. The fact that high CSF concentrations of β -lactam antibiotics may cause neurologic effects, including seizures, should be considered. Because cefoperazone and sulbactam are both removed from the circulation by hemodialysis, these procedures may enhance the elimination of the drug from the body if overdosage occurs in patients with impaired renal function.

Presentation

1 vial or 10 vials are packed in a folded paper box.

Storage Condition

Store below 30°C. Balance solution should be discarded.

Shelf Life

30 months from the date of manufacture if kept as recommended.

Registration number

Cefpar SB Injection 1 g - MAL09062135AZ

Name and Address of Manufacturer

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